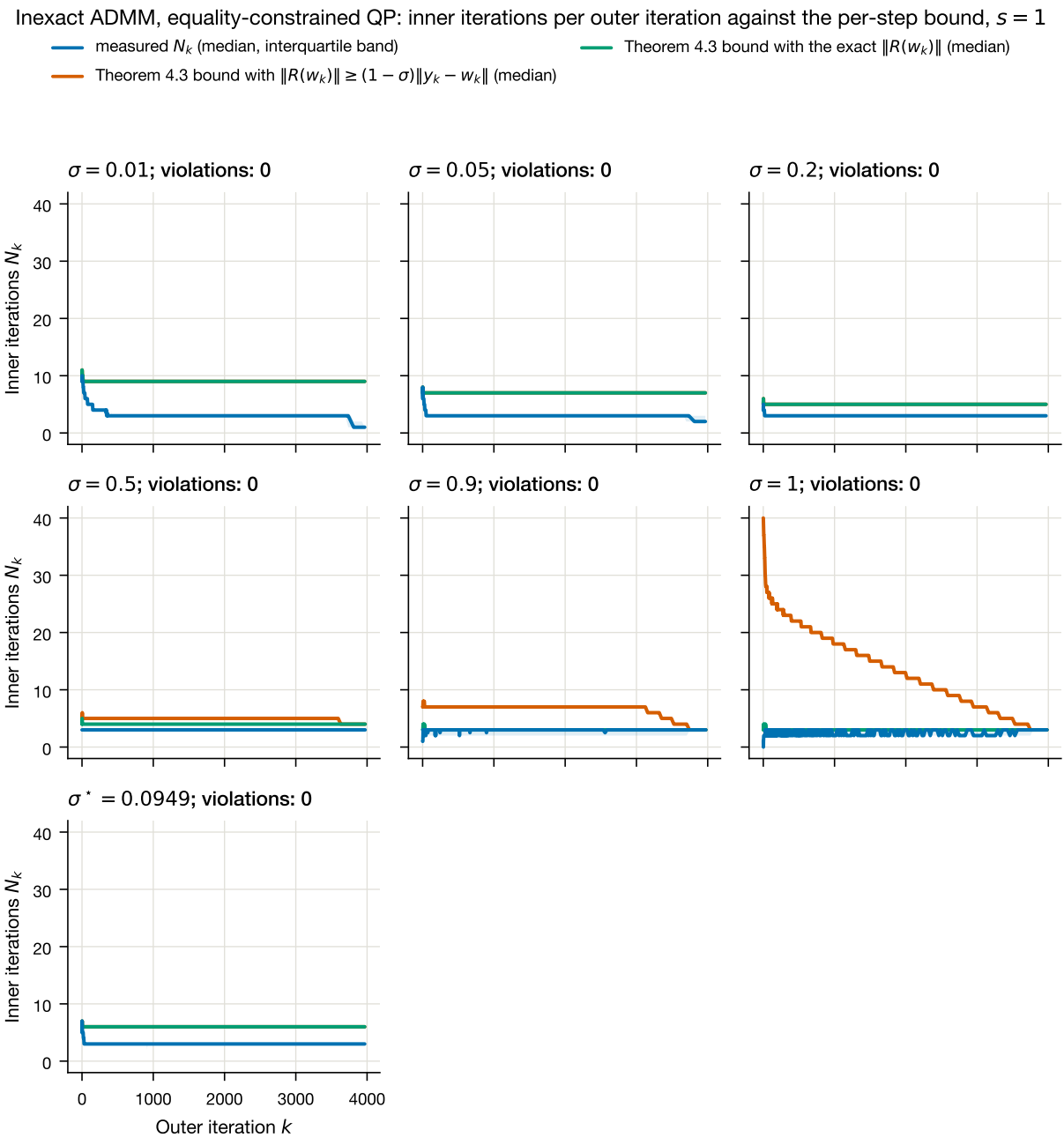

Inexact ADMM, equality-constrained QP: inner iterations per outer iteration against the per-step bound, $s = 1$
measured N_k (median, interquartile band)
Theorem 4.3 bound with the exact $\|R(w_k)\|$ (median)
Theorem 4.3 bound with $\|R(w_k)\| \geq (1-\sigma)\|y_k - w_k\|$ (median)
$\sigma = 0.01$; violations: 0
$\sigma = 0.05$; violations: 0
$\sigma = 0.2$; violations: 0
$\sigma = 0.5$; violations: 0
$\sigma = 0.9$; violations: 0
$\sigma = 1$; violations: 0
$\sigma^\star = 0.0949$; violations: 0
Inner iterations N_k
0
10
20
30
40
0
1000
2000
3000
4000
Outer iteration k